

Power Delivery Network (PDN) Modeling for Backside-PDN Configurations With Buried Power Rails and μ TSVs

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Abstract—In this article, a power delivery network (PDN) modeling framework for backside-PDN configurations is presented. A backside-PDN configuration contains dense microthrough silicon vias (μ TSVs) and power/ground metal stack on the backside of the die. This approach separates the PDN from a conventional signaling network of the back-end-of-the-line (BEOL) and improves power integrity and core utilization. We benchmark this technology with conventional front-side BEOL PDN configurations. Owing to the lower resistivity compared with Cu metal lines for advanced technology nodes, we use ruthenium (Ru)-based buried power rail for PDN modeling. Our analysis shows that the steady-state IR-drop reduces by more than 4 $\times$ in the backside-PDN configuration, and a simultaneous switching noise analysis shows a significant reduction in transient droops. The framework results are validated with a place-and-route (P&R)-based physical implementation flow. We quantify the area improvement in the actual flow and observe 25%–30% improvement in the backside-PDN configuration. From a PDN modeling framework, the PDN results follow a trend similar to the ones obtained from the block-level P&R of the given configurations. Moreover, we investigate the impacts of package-to-die interconnect pitch, metal–insulator–metal cap density, and input pulse on the PDN performance. In addition, we perform thermal modeling to analyze the thermal implications of the backside-PDN configuration. From a thermal modeling perspective, there is negligible influence from a dielectric bonding layer in the backside-PDN configuration.

Index Terms—Interconnects, IR-drop, physical design, power delivery, transient droop.

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I. INTRODUCTION

THERE is an increasing interest in modular-level heterogeneous integration of multifunctional dice into a single package. A single package may include high-bandwidth memory with GPUs [1], FPGAs with server processors [2], high-performance GPUs with general-purpose CPUs [3], and so on. Despite the scaling of supply voltage in recent device technologies [4], these high-performance integrated modules inevitably lead to higher current demand and increased power density [5]. As a result, power delivery in high-performance digital systems is an increasingly difficult challenge [6]. While there are a number of solutions to improve the efficiency and reduce the footprint of the active portion of a power delivery network (PDN) [7], [8], one has to rethink the on-die PDN design to achieve the best out of both scaling trends and innovative packaging solutions. Specifically, scaling device technology poses several challenges. The resistivity and resistance of conventional metal layers and intermetal vias are increasing rapidly with advanced technology scaling [9], [10], while the PDN noise margins are getting stringent [11]. Moreover, the power consumption of different computing blocks is increasing significantly [12], [13]. Power supply noise (PSN) negatively impacts the system performance; PSN induces clock jitter that exacerbates the performance of a computing block [14]. Modern processors can create nanosecond-level voltage droops that require different circuit techniques to ensure reliability [15], [16]. Hence, all these challenges call for PDN analysis in the predesign phase to identify the pros and cons of emerging materials, technologies, and topologies.

Backside-PDN configuration attempts to tackle some of these challenges by separating the on-die PDN from the conventional back-end-of-the-line (BEOL) [17]. This approach is a complete redesign of existing architectures in that both sides of the silicon have metallization layers. Moreover, alternative metallization is considered for the bottommost metals to tackle the resistivity challenges. Therefore, in this article, based on our prior PDN modeling techniques [8], [18], we develop a framework to analyze PSN in the backside-PDN configuration. Furthermore, using this framework, we benchmark the backside-PDN configuration with respect to a conventional

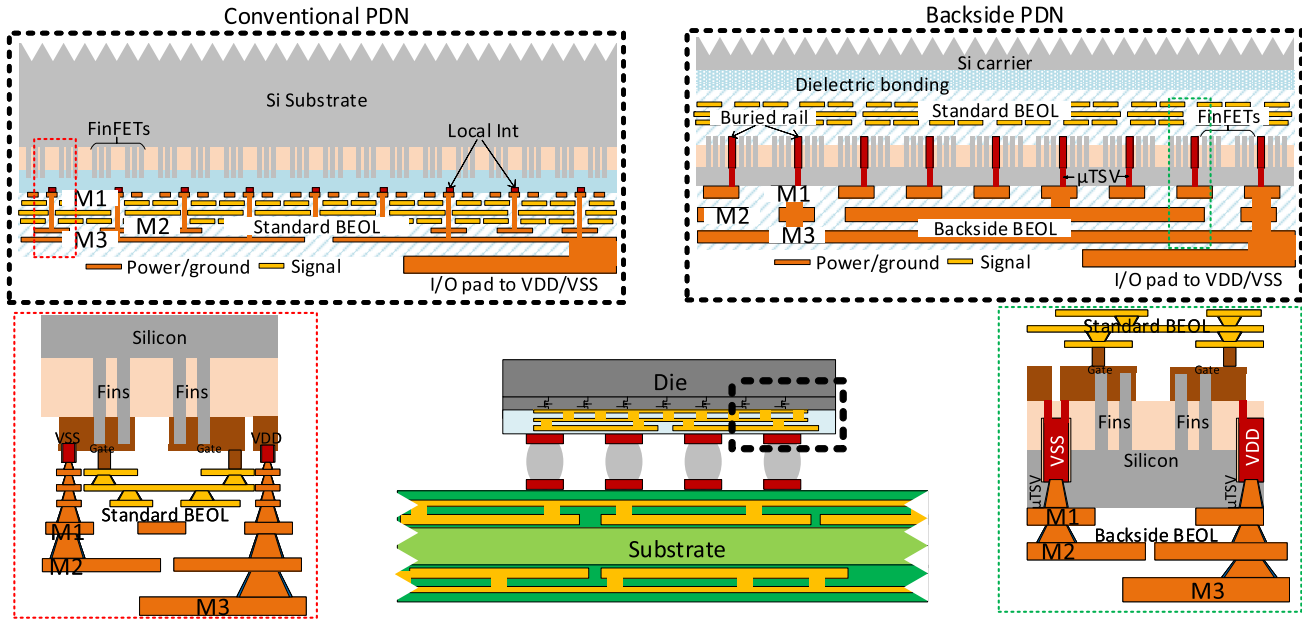


Fig. 1. Die placement and metal configurations for a conventional front-side PDN configuration and backside-PDN configuration.

BEOL PDN to identify unique opportunities as well as limitations of this approach.

This article is organized as follows. In Section II, we introduce the differences between the backside-PDN and conventional front-side PDN configurations. In Section III, we evaluate the power delivery performance of the backside-PDN configuration. We present results for different power maps and compare modeling results with the physical design results. In Section IV, we perform a design space exploration; we analyze the impacts of package-to-die interconnection pitch, input pulse, and capacitor density on the PDN performance. In addition, we investigate the thermal implication of dielectric bonding for the backside-PDN configuration.

II. MODELING FRAMEWORK AND SPECIFICATIONS

A. Simulation Configurations

In a conventional PDN approach, as shown in Fig. 1, die VDD/VSS I/Os interconnect to the global metal PDNs in the BEOL. Next, the global PDN connects to the local PDN (bottom metal layers and local interconnects) using intermetal vias [19]. Fig. 1 also shows the backside-PDN configuration that we explore in this article. Fig. 1 shows the die placement on the package and a detailed segment of the die highlighting the supply (VDD or VSS) I/Os and the PDN. There are two types of metallizations in this configuration. The conventional BEOL is located on the front side of the die and is directly connected to the front-end-of-the-line (FEOL). This front-side BEOL interconnect network is primarily dominated by the signaling network. In the proposed approach, the backside metallization of the die is dedicated to the PDNs. In addition, this configuration has a buried power rail (BPR) network within the FEOL to locally supply power to active circuits. BPR is interconnected to the backside PDN using microthrough silicon vias (μ TSVs) [20], [21].

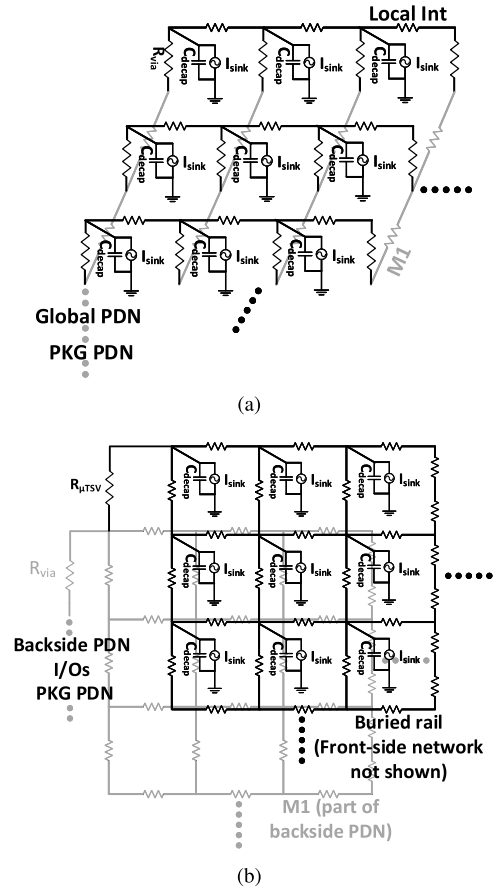


Fig. 2. On-die PDN structure for (a) conventional interleaved BEOL PDN configuration and (b) meshed backside-PDN configuration.

We consider three separate PDN domains in the modeling [8], [22]: on-die, package, and board-level PDNs. In the backside-PDN configuration, we assume a redesign of the on-die PDNs. Package and board domains remain similar in

TABLE I
PDN SPECIFICATIONS FOR DIFFERENT CONFIGURATIONS

Parameters	Conventional PDN	Backside PDN
No. of metal layers	3	3
PDN metal	Local Int: Cu, M1: Cu, M2: Cu	BPR: Ru, M1: Cu, M2: Cu
PDN metal resistance ($\Omega/\mu\text{m}$)	Local Int: 500, M1: 1, M2: 1	BPR: 50 (170 nm height, 36 nm width [23]), M1: 1, M2: 1
μ TSV Diameter/Height/Resistivity (nm/nm/n Ωm)	N/A	50/500/80
Via resistance (Ω/via)	Local Int-M1: 160, M1-M2: 2	BPR-M1: 24, M1-M2: 2
Die-to-package bumps	Diameter: 70 μm , Height: 140 μm , Pitch: 140 μm	Diameter: 20 μm , Height: 40 μm , Pitch: 40 μm
On-die decoupling cap (nF/mm ²)	1.8	1.8
Package effective decap R/L/C (m Ω /pH/ μF)	541.5/220.7/52	541.5/220.7/52
Package resistance/inductance (m Ω /mm/pH/mm)	1.2/24	1.2/24
PCB decap R/L/C ($\mu\Omega$ /nH/ μF)	166/19.54/240	166/19.54/240
PCB resistance/inductance ($\mu\Omega$ /pH)	166/21	166/21

both configurations. Fig. 2 shows the structure of an on-die PDN. The conventional PDN (within BEOL) is an interleaved structure, as shown in Fig. 2(a), whereas the backside PDN is a meshlike network. Fig. 2(b) shows the on-die PDN for a backside-PDN configuration. The bottom two metal layers in each configuration are connected by different via resistances: dense intermetal via stack in the conventional BEOL configurations and μ TSVs in the backside configurations. Moreover, the topmost metal layer is connected to the package PDN by C4 bumps and copper pillars in the conventional BEOL PDN and backside PDN, respectively.

B. Specifications

In each configuration, we consider three metal layers (local interconnect/BPR, M1, and M2) for the PDN. The specifications are tabulated in Table I. We consider high aspect ratio Ru-based bottommost metal layer for the backside-PDN configurations [23], [24]. For advanced technology nodes, Ru provides a significant reduction in resistivity compared with other conventional metal options (Cu, Co, and so on) [10]. Moreover, in the backside-PDN configuration, the package-to-die bumps are denser compared with the conventional PDN; denser bumps improve the distribution of current. We use similar inductance values for package traces, microbumps, solder bumps, and so on, as in our prior PDN modeling effort [8]. In addition, the intermetal via resistance for bottom metals is $\sim 7\times$ smaller in the backside configuration compared with similar connections in the conventional BEOL PDNs [10], [25]. This reduction can be attributed to the via stack through

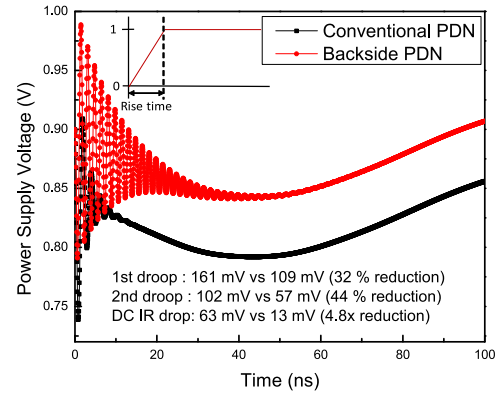


Fig. 3. PSN results for uniform power map.

the signaling network in a conventional BEOL [19]. We consider both uniform and hotspot-based power maps for the PDN analysis. In uniform power map analysis, total die power is uniformly distributed across the die. In hotspot power density maps, some areas of the die consume significantly more power than the rest of the die. For the uniform power map case, we consider 74.49-W/cm² [5] power density. Unless otherwise specified, we use this power for uniform power map analysis throughout this article. We assume a 5 mm $\times$ 5 mm die with 0.9-V rail voltage throughout this article. We consider two types of on-die decoupling capacitors: die-level MOS caps and metal-insulator-metal (MIM) capacitors (caps) connected to the top metal layers.

C. Adaptive Meshing

We use domain-specific adaptive meshing for PDN modeling; we use different grid sizes for on-die, package, and board PDNs. For hotspot power density analysis, we use within-die adaptive meshing technique. In our prior PDN modeling, we have used package-to-die bump granularity for meshing [8]. In within-die adaptive meshing, we use denser grids in the hotspot regions while using coarse grids for the rest of the die. For example, if we have a 100 μm $\times$ 100 μm hotspot in a 5 mm $\times$ 5 mm die for the front-side PDN configuration, we use 1- μm grid for the hotspot and 140- μm grid for the rest of the die. Using a 1- μm grid for the whole die would yield 25-M nodes for a single layer of the PDN. On the contrary, using a 140- μm grid would compromise on the accuracy of the PDN model. The adaptive meshing technique reduces the number of total mesh elements while performing a fine-grained PDN analysis for critical die blocks.

III. PDN BENCHMARKING

In this section, we present the PSN results for both configurations. We exclude MIM caps from the analysis in this section.

A. Uniform Power Density Maps

We explore the step response-based simultaneous switching noise for both configurations using a 400-ps rise time. Fig. 3

summarizes the results. We report the improvement in each noise droop for the backside-PDN configuration with respect to its conventional front-side counterpart. From the dc IR-drop analysis, the backside PDN provides more than $4\times$ reduction in PSN. This reduction can be attributed to the denser PDN and denser package-to-die bumps in such a configuration. Examining the inductive noise, we observe a significant 32% and 44% reduction in the first and second droop noises, respectively. Only the on-die PDNs are modified between two configurations. The unmodified package PDN is mostly inductive, and hence, for this uniform power map case under consideration, we do not observe as much reduction in the first droop noise as the dc IR-drop noise.

B. Hotspot Power Densities

The uniform power map case emulates the worst case scenario where all the die nodes are switching simultaneously. However, this is an average power map across the die. The microscale circuit blocks have a higher power density compared with an average power map [8]. Moreover, we have stated previously that owing to higher densities, advanced technology nodes, such as 7 and 3 nm, will consume significantly higher power [13]. We emulate such an aggressive microscale power map in Fig. 4(a). While we keep the die size unchanged as $5\text{ mm} \times 5\text{ mm}$, we assume that 0.3 W is being consumed in a $100\text{ }\mu\text{m} \times 100\text{ }\mu\text{m}$ region. Moreover, we assume that the rest of the die is not consuming any power. For simplicity, we begin with one such hotspot and observe 32- and 130-mV peak IR-drop in the backside-PDN and conventional BEOL PDN configurations, respectively. This single hotspot simulation is an emulation of a standalone computing block [13], [17]. To emulate a more realistic scenario, we add four additional hotspots surrounding the first one. This equates to a total die power of 1.5 W . This case emulates multiple cores or computing blocks running in parallel. For the five hotspots case, we observe 50- and 240-mV peak IR-drop in the backside-PDN and conventional BEOL PDN configurations, respectively. Fig. 4(b) summarizes these results. For the backside-PDN configuration compared with its conventional counterpart, the single hotspot and five hotspots cases provide $4\times$ and $5.4\times$ reductions in PSN, respectively. The Ldi/dt noise reflects the impact of both inductive and resistive components of a network. Since the hotspot power density maps create significantly higher dc IR-drop in the conventional BEOL PDN configurations, unlike the uniform power map case, we observe a similar improvement in both steady-state and transient analysis results for the backside-PDN configuration. Fig. 4(c) shows the results for five hotspots case with a total power of 18.6 W . This case is a combination of the uniform power map and the high-density hotspot power map under consideration. In this analysis, the first droop noise is $2.5\times$ lower in the backside-PDN configuration compared with the front-side PDN case. However, similar to other case studies with different power maps, the second and third droops show a similar greater than $4\times$ reduction. In Fig. 4(b) and (c), the total power consumption is 1.5 and 18.6 W , respectively. For higher power consumption, the package and board losses

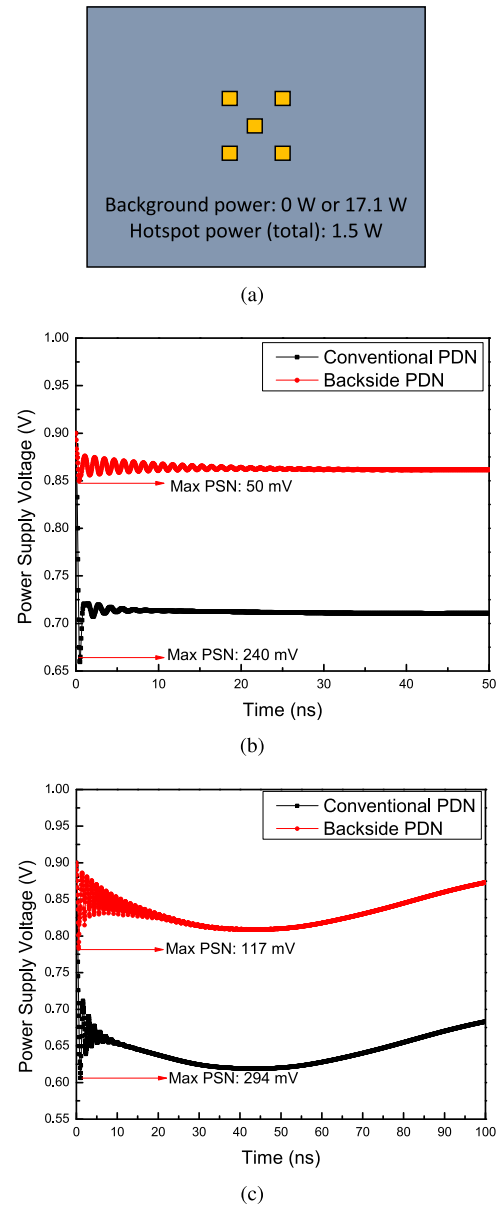


Fig. 4. (a) Power map with five adjacent hotspots. (b) PSN results for the hotspot power map with zero background power. (c) PSN results for the hotspot power map with 17.1-W uniform background power.

are significantly higher, and hence, compared to Fig. 4(b), we observe an increase in the second and third droop noises in Fig. 4(c). For a zero background power case, only a few on-die nodes are switching as opposed to all on-die nodes in a nonzero background power case. Hence, we do not observe greater than $4\times$ reduction in the first droop noise, as we observe in Fig. 4(b).

C. Physical Design Results

We perform physical design implementation for both configurations for an RISC-V architecture. Table II summarizes the results. For the conventional BEOL PDNs, we implement PDNs of different densities. We use contacted poly pitch (CPP) to reflect the scaling of technology nodes [17] and, hence, PDN density. Furthermore, we implement conventional BEOL PDN with BPR, backside PDN with standard

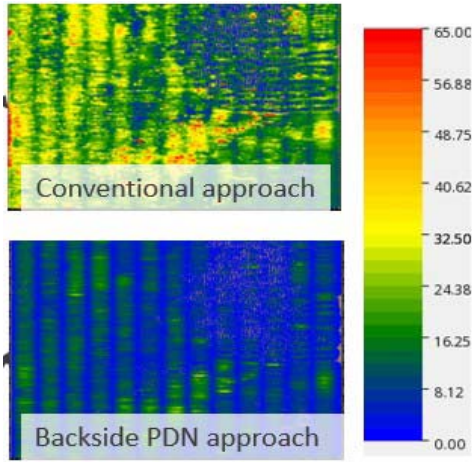


Fig. 5. Physical design results for different PDN configurations.

TABLE II
SUMMARY PHYSICAL DESIGN RESULTS FOR
AN RISC-V ARCHITECTURE

Technology	PDN density	Area (μm^2)	Peak IR-drop
conventional BEOL PDN	8 CPP	8594	4x
	16 CPP	7365	7.5x
	24 CPP	6874	9x
conventional BEOL PDN + BPR	48 CPP	6874	4x
Backside PDN excluding BPR	32 CPP	6446	4x
Backside PDN + BPR	8 CPP μ TSV	5926	1x

power rails, and backside PDN with BPR. After placement and routing, we observe a 25%–30% area reduction in the backside-PDN configurations. We normalize the peak IR-drop in each configuration to the IR-drop value for the backside-PDN configuration with BPR. From physical implementations, we observe that each metric has a minimum 4 $\times$ reduction for a backside-PDN configuration with BPR. In Table III, we compare the PDN modeling results with the physical implementation results. The conventional BEOL PDN is a dense 8 CPP design. The backside-PDN configuration is an 8 CPP design with μ TSVs. From physical design results, we observe better core utilization in the backside-PDN configurations. Fig. 5 shows the extracted IR-drop results from physical implementation for both configurations. Although the physical design is an exact architectural implementation, whereas the PDN modeling framework in this article is abstract modeling from high-level PDN parameters, e.g., the number of metal layers, dimensions of PDN metals, and decoupling cap densities, both in steady-state IR-drop and transient Ldi/dt noise analyses, we observe similar trends between physical implementation and PDN modeling results. Owing to the generalization in the PDN modeling with larger die size and inclusion of the package and the board PDN, we observe some discrepancy between the results of the physical design and the PDN modeling framework.

IV. SENSITIVITY ANALYSIS

In this section, we show results from design space explorations to determine the limits and benefits of the backside-PDN configuration.

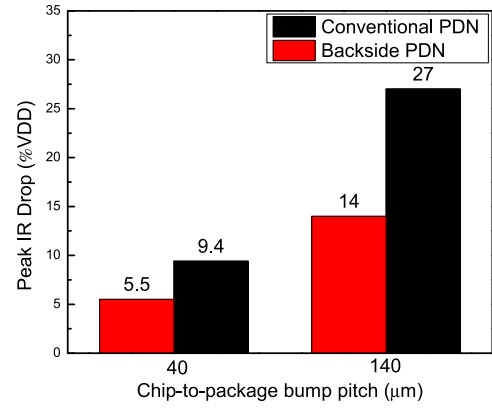


Fig. 6. Peak IR-drop comparison for different package-to-chip bump pitches.

A. Chip-to-Package Interconnection

Throughout this article, we assume 40- μm pitch die-to-package interconnections for backside PDNs and 140- μm pitch for conventional BEOL PDNs. In Fig. 6, we report results for varying pitches in each configuration. We report the maximum transient Ldi/dt noise for each variant. For each pitch value, backside PDNs provide almost 2 $\times$ reduction in PSN compared with its conventional BEOL counterpart. This reduction can be attributed to the denser PDN along with lower via resistance for the lowest metal layers. This essentially means better performance in the backside PDNs. Moreover, we observe that the scaling I/O pitch can have a significant impact regardless of the configuration, and denser power/ground (P/G) I/Os are favorable in both cases. For both configurations, compared with 140- μm pitch, 40- μm pitch interconnections provide 2 $\times$ improvement in the PSN. An assembled die has successively higher resistance metal traces from board to die-level PDNs. As such, between the package PDN and the die PDN, the package PDNs have lower resistance and, hence, help spread the current. Denser P/G bumps enhance this spreading in the package level. Hence, we observe this improvement in the PSN with respect to bump pitch reduction.

B. Impact of Input Pulse

In this analysis, we evaluate the step response by varying the rise time of the input current load. We assume the uniform power map discussed in this article. We sweep the rise time from 200 ps to 1 ns. Fig. 7 shows the results for these different rise times. For reference, we also report the step response for the conventional BEOL PDN with 1-ns rise time. As expected, with increased rise time, the PSN reduces in the backside configuration. However, between a backside-PDN switching with 400-ps rise time and a conventional BEOL PDN switching with 1-ns rise time, we observe a similar supply noise. Hence, the backside-PDN configuration enables faster switching compared with a conventional front-side PDN configuration.

C. MIM Decoupling Cap Density

One additional advantage of the backside-PDN configuration is to have denser MIM caps connected to the top-level

TABLE III
SUMMARY OF PSN RESULTS FROM PHYSICAL DESIGN AND PDN MODELING

Configuration	PDN density	Target density (%)	Peak IR-drop from physical design	Peak steady-state IR-drop from PDN modeling	Peak I_{di}/dt noise from PDN modeling
conventional BEOL PDN	8 CPP	65	4x	4x	5x
Backside PDN	8 CPP μ TSV	87	1x	1x	1x

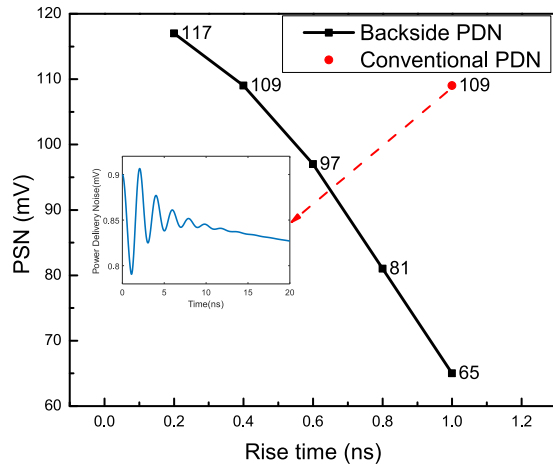


Fig. 7. Impact of rise time variation on step response for backside-PDN configuration. The red line shows the step response result for the conventional BEOL PDN with 1-ns rise time.

metal layers. This is facilitated by the separation of signaling layers from the PDNs of the dice. Throughout this article, we exclude the MIM caps from the analysis. In this section, we investigate the impact of the MIM cap density on PSN, as shown in Fig. 8(a). We use a uniform power map and step response-based simultaneous switching noise analysis for this analysis. We vary the MIM density from 0 to 50 nF/mm². The 0-nF/mm² MIM density corresponds to the backside-PDN analysis results shown in Fig. 3. As we increase the MIM cap density, the transient droop reduces. However, beyond a certain cap density, the PSN does not improve further. From the figure, we can see that beyond 10 nF/mm², the second droop begins to dominate the PSN. Fig. 8(a) (inset) shows the noise profile for a 50-nF/mm² MIM cap density. This is a representative of noise profiles with MIM cap densities greater than 10 nF/mm².

In the second part of this analysis, we explore a different input excitation for the PSN analysis. We analyze simultaneous switching noise for an input pulse with 1-GHz frequency. The input has 400-ps rise time, 200-ps conduction time, and 400-ps fall time, respectively. Fig. 8(b) summarizes the results for this analysis. We consider three different MIM cap densities (0.5, 5, and 50 nF/mm²). Evidently, as we increase the MIM cap density across the die, both supply noise and supply noise fluctuation reduce significantly. As we increase the MIM cap density from 0.5 to 5 nF/mm², the peak-to-peak fluctuation of the supply voltage reduces by more than 3 $\times$. A higher cap density may not improve the first droop, and however, it helps reduce the high-frequency noise ripple.

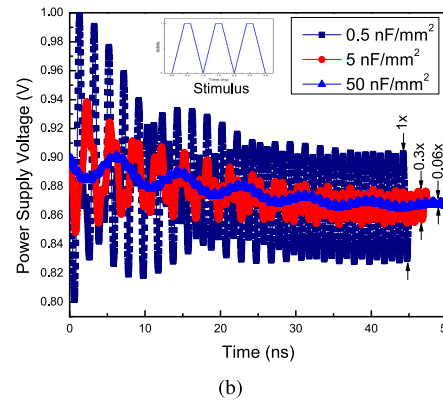
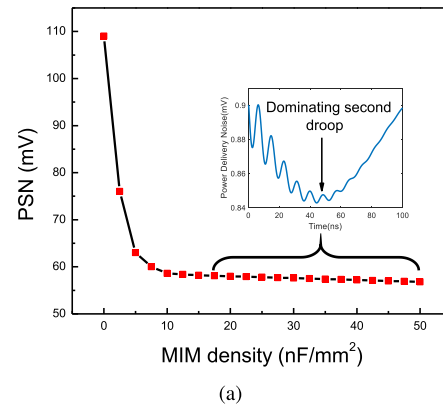


Fig. 8. (a) Step response results for different MIM densities and (b) supply noise for 1-GHz pulse input.

D. Thermal Implications of a Backside-PDN Configuration

The backside-PDN configuration uses a dielectric bonding of the active layers with a carrier wafer. Compared with a conventional front-side PDN configuration, this bonding layer increases the junction-to-ambient thermal resistance. We use our thermal modeling framework [5] to evaluate the impact of this additional layer on the thermal performance of the backside-PDN configuration. For this analysis, we assume a 1 cm $\times$ 1 cm die with 74.49-W power. Although the dielectric bonding layer is typically a 1 μ m $\sim$ 2 μ m, we simulate for up to 20- μ m dielectric bonding layer in the backside-PDN configuration. We assume that the dielectric bonding layer has a thermal conductivity of 0.9 W/m-K, which is similar to the thermal conductivity of SiO₂. Moreover, we assume an air-cooled heat sink with 0.218- $^{\circ}$ C/W case-to-ambient thermal resistance.

According to our analysis, the additional dielectric bonding layer in the backside-PDN configuration has a negligible impact on the temperature distribution of the dice.

The conventional front-side PDN configuration and the backside-PDN configuration have a maximum junction temperature of 79.5 °C and 79.6 °C, respectively.

V. CONCLUSION

In this article, we present a PDN modeling framework for the backside-PDN configurations. The backside-PDN configurations are similar to the double-side processed dice with signaling network and PDN on either side of the FEOL. Owing to the denser PDN and new materials for the bottommost metal, this configuration provides significant improvement in power supply voltage. We use a uniform power map to emulate a real computing block and a nonuniform power map to emulate futuristic computing blocks at advanced technology nodes. For both power maps, we observe greater than 4 $\times$ reduction in PSN in the backside-PDN configurations relative to conventional BEOL counterparts. We use the physical implementation of an RISC-V architecture to validate our modeling results. Our physical design shows a 25%–30% area improvement in the backside-PDN configuration compared with the conventional BEOL configurations. Our package-to-die bump pitch analysis shows at least 2 $\times$ performance improvement for the backside-PDN configuration over a conventional counterpart. We sweep the rise time of the input pulse. We observe that the backside-PDN configuration with 400-ps rise time provides a similar noise profile of a conventional BEOL configuration with 1-ns rise time. Moreover, based on our assumptions, we observe that an MIM cap density greater than 5 nF/mm² does not improve the first droop noise further, and however, more MIM caps can reduce high-frequency ripple for a given input pulse. Despite the inclusion of a thermally resistive bonding layer, thermal modeling results indicate that the backside-PDN configuration has similar temperature distribution as a front-side PDN configuration.

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